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Grover-Inspired Evidence Selection for Document Retrieval: Relevance Constraints, Failure Analysis, and Resource Accounting

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ABSTRACT Selecting a fixed number of documents requires more than identifying relevant passages: the selected passages should also be useful together. This article examines a Grover-inspired selection pipeline that combines classical retrieval, cross-encoder relevance scores, and ideal amplitude amplification simulated on a central processing unit. A base selector chooses eight documents from fifteen candidates. A constrained refinement preserves the first five results and searches an expanded pool for a less redundant three-document tail. We synthesize completed evaluations over 1,271 test queries from scientific, nutrition-related, and financial collections, and extend their analysis without collecting new benchmark rankings. The reported normalized discounted cumulative gain at rank five is 0.6878 on SciFact, 0.3689 on NFCorpus, and 0.3575 on FiQA. Refinement decreases mean redundancy by 3.21 and 2.63 percent on the first two collections, while relative recall changes by -0.93 and +1.95 percent. FiQA retains the earlier base configuration; its separate refinement failure remains disclosed. Candidate-coverage bounds, query-level losses, seed stability, and request-level costs reveal effects that aggregate metrics can hide. Matched greedy and exact refiners attain the same reported aggregate quality, and neural inference dominates measured online time. A separate completed resource study characterizes reversible constraints and coefficient precision, including explicit simulation limits. The evidence supports a constrained relevance–redundancy tradeoff and reproducible resource accounting; it establishes no physical quantum acceleration or improvement in generated answers.

INDEX TERMS Amplitude amplification, cross-encoders, document retrieval, evidence selection, quantum algorithm simulation, reproducibility.

I. INTRODUCTION

AN evidence budget makes document retrieval a selection problem as well as a ranking problem. A system returning eight passages must decide whether a ninth candidate supplies useful additional information or merely repeats material already retrieved. Independent relevance estimates do not, by themselves, resolve this interaction. The distinction matters in retrieval-augmented generation, where selected passages occupy a limited context window and determine which evidence a generator can consult [1]. It also matters in ordinary document search: a compact list can contain individually plausible results while providing little variety.

Lower similarity between documents does not necessarily improve an answer. A passage that resembles another may contain the only correct support for a claim, while a dissimilar passage may be irrelevant. Relevance, redundancy, evidence coverage, and downstream answer quality must therefore remain separate quantities. The present work studies judged retrieval effectiveness and an embedding-based redundancy proxy. It neither substitutes that proxy for human evidence judgments nor attributes unmeasured generation outcomes to the retrieval stage.

Quantum search offers a useful setting in which to make these distinctions explicit. Grover search and amplitude amplification give an oracle-query improvement under their stated access assumptions [2], [3]. A document pipeline, however, first constructs representations, retrieves candidates, computes relevance scores, and encodes an objective. A later amplitude-amplification step does not eliminate these operations. Simulating amplitudes on a classical computer adds

work of its own. Its measured latency cannot be identified with the complexity of an ideal black-box quantum oracle.

Our starting point is the accepted Qubit Chat conference manuscript [4]. That work reports controlled retrieval comparisons, an L6-guided Grover Adaptive Search (GAS) selector, a protected-prefix refinement, mathematical bounds, and a bounded native-oracle resource study. Those contributions and their experimental results are prior work. This journal article revisits the same completed evidence to answer questions that require finer analysis than a compact conference presentation permits: where relevant documents disappear from the candidate pipeline; how many queries improve or deteriorate despite a favorable mean; whether five stochastic seeds yield meaningfully different outcomes; and which measured stages account for the complete request cost.

The base policy selects eight documents from a hybrid shortlist of fifteen. A cross-encoder supplies the relevance component, while clipped pairwise embedding cosine measures redundancy. The subsequent refinement freezes the baseline's first five results, expands the candidate pool within the original dense top 64, and searches for a new three-document tail. Tail-score constraints restrict the search, and acceptance requires a strict decrease in the unquantized redundancy measure. Preserving the prefix fixes normalized discounted cumulative gain at rank five (nDCG@5), but it leaves recall and lower-ranked relevance free to change. This asymmetry is central to interpreting the evidence.

Three completed public test collections provide the empirical basis: SciFact, NFCorpus, and FiQA. SciFact and NFCorpus use the protected refinement; FiQA's reported row uses its earlier base policy. The article retains this distinction rather than combining metrics from different configurations into a single favorable row. The later failed FiQA refinement and NFCorpus confirmation exception remain part of the scientific record. Because earlier test outcomes informed method development, the follow-up tests support exploratory evaluation rather than an untouched confirmatory assessment.

The journal extension adds four analyses. We measure candidate-pool coverage and the best Recall@8 attainable within each saved pool, then examine query-level gains, losses, and ranking variation across seeds. We also reconstruct request-level latency to compare the costs of relevance scoring, base selection, and refinement. Finally, we connect these diagnostics to the recorded precision and resource grid and map the reported values to sealed artifacts. No new benchmark rankings, neural inference, or dataset collection are introduced.

The resulting claim is specific: constrained set selection can reduce measured redundancy while keeping a selected ranking prefix unchanged. Its recall consequences depend on the collection and on the relevance model's errors. Agreement with matched classical optimizers identifies a property of the formulation rather than a uniquely Grover-derived effectiveness gain. The resource evidence characterizes an implementation and its limits; it does not establish an operational quantum advantage.

The remainder of the article situates the formulation in retrieval and quantum optimization, specifies its algorithms and evaluation protocol, reports the recorded effectiveness results, and develops the journal-specific diagnostics. A separate resource section distinguishes compact simulation from native reversible-oracle synthesis. The discussion connects these findings to their validity limits and to experiments needed before stronger deployment claims could be made.

II. RELATED WORK

A. RELEVANCE ESTIMATION AND EVIDENCE-SET SELECTION

Lexical and dense retrieval capture different aspects of relevance. BM25 estimates term-based matching [5]; sentence encoders produce reusable document and query vectors [6]. Reciprocal-rank fusion (RRF) combines rankings without assuming that the underlying scores share a calibrated scale [7]. Here, these methods define candidate pools and classical comparisons. Their costs remain part of the retrieval pipeline when we report complete request time.

Cross-encoders jointly represent a query and a passage, enabling richer relevance interactions than independent vector encoding [8], [9]. The MiniLM embedding encoder and the MiniLM cross-encoder used here share a model-family name but have different roles [10]. The first supplies normalized vectors; the second supplies query–passage logits from MS MARCO passage-ranking training [11]–[13]. A logit is neither a calibrated probability of relevance nor a judgment label. Improving logits therefore does not guarantee higher judged recall.

Alternative neural architectures include dense passage retrieval [14], late-interaction ranking [15], and sequence-to-sequence relevance models [16]. Generative ranking agents further broaden the design space [17]. Contemporary reranker evaluations examine architecture, training overlap, generalization, and computational cost [18]. These works motivate stronger future comparisons, but their published results cannot be inserted into our tables as if they were local experiments. The evaluated neural controls here are specifically the pinned L2 and L6 cross-encoders on their declared pools.

Recent empirical RAG evaluation also separates indexing, retrieval, reranking, similarity, generation quality, and system cost [19]. We use the same separation to define our measurement boundaries. The saved experiments address retrieved evidence and selection resources; assessing generated answers requires further measurements.

Maximal marginal relevance (MMR) chooses documents one at a time, balancing relevance against similarity to earlier choices [20]. Our objective evaluates the average pair similarity of a complete set of fixed size. Determinantal point processes provide a probabilistic model for diverse subsets [21]. They supply literature context here and are not evaluated in these runs. Table 1 separates the model of document interactions from the mechanism used to search its objective.

TABLE 1. Methodological positioning. "Measured" identifies local controls; literature-only rows do not assert an empirical comparison.

Approach	Relevance signal	Treatment of document interactions	Relationship to this evaluation
BM25 / dense / RRF [5]–[7]	Lexical, vector, or fused ranks	Candidate scoring; no joint redundancy objective	Measured classical policies and upstream features
L2/L6 cross-encoders [9], [13]	Joint query–passage relevance	Scores documents independently	Measured on declared dense or hybrid pools
MMR [20]	Estimated relevance	Sequential novelty penalty	Measured cosine/L6 variants with frozen tuning
Determinantal selection [21]	Quality-weighted similarity model	Probabilistic subset diversity	Literature context; not an evaluated baseline
Generative reranking [17], [18]	Language-model ranking judgments	Model-dependent listwise interactions	Literature context; no local measurements
Classical matched optimizers	Identical saved L6 features	Same set objective and admissibility rules	Greedy, exact float/integer, and random controls
GAS [22]	Encoded polynomial objective	Adaptive improvement search	Adopted search framework; no new GAS algorithm
Protected-prefix policy [4]	Original and additional L6 logits	Fixed head, constrained tail, pair redundancy	Prior formulation; expanded analysis in this article

B. QUANTUM SEARCH AND REVERSIBLE RESOURCE ACCOUNTING

Amplitude amplification raises the probability of a marked subspace. It gives equally marked documents no intrinsic relevance ordering [2], [3]. Randomized schedules address unknown marked-state counts [23]; minimum finding updates an incumbent threshold [24]. GAS extends these ideas to constrained polynomial binary optimization [22]. Our subset objective adopts that framework and explicitly represents interactions among selected documents.

Recent GAS work studies schedule efficiency and the distinction between asymptotic query complexity and its constant factors [25]. Application-specific analyses also examine initialization, Grover rotations, and gate construction [26]. These studies are relevant because an oracle call is not a unit of physical execution time independent of the encoded problem. Our measured oracle incorporates cardinality, tail-score thresholds, signed improvement, and uncomputation; the compact simulator uses a different representation with a different cost boundary.

Fixed-weight state preparation can exploit structure, as illustrated by deterministic Dicke-state constructions [27]. A structural preparation result is not a measured circuit resource for software that uses generic state preparation. Similarly, matrix-product-state simulation has costs that depend on entanglement [28]; a successful small simulation does not certify feasible execution of all larger circuits. Broader quantum-algorithm surveys emphasize access models and implementation assumptions [29]. These distinctions inform both the resource measurements and the language of our conclusions.

C. EVALUATION PRACTICE AND EXTENSION OVER PRIOR WORK

BEIR supplies heterogeneous collections and standardized retrieval tasks [30]. SciFact, NFCorpus, and FiQA contribute scientific claims, nutrition-related search, and financial questions, respectively [31]–[33]. Their different judgment distributions make absolute recall values difficult to compare without considering candidate coverage and the number of relevant documents per query. The new coverage analysis addresses that interpretive issue within each collection.

Gain-based ranking evaluation and independent implementations support metric consistency [34], [35]. Paired inference must use queries as observations, preserve the pairing between methods, and account for the family of comparisons [36]–[38]. The present article retains the recorded statistical families and adds descriptive diagnostics rather than retroactively changing the hypotheses. Compared with the conference manuscript [4], its new empirical content consists of secondary analyses of saved outcomes, not additional test collections or a newly tuned retrieval policy.

D. APPLICATION-SPECIFIC DECISION PIPELINES

Task-specific decision pipelines also appear in blockchain mining: trust scoring [39], [40], VIKOR-based ranking [41], reputation and consensus mechanisms [42], [43], and miner selection or classification [44], [45]. Other application studies examine multimodal waste detection [46], smart-contract repair [47], financial-fraud detection [48], plant-disease recognition [49], traffic forecasting [50], and obscured-fruit classification [51]. These studies provide application context; their tasks and reported results are separate from our document-retrieval evaluation.

III. PROBLEM FORMULATION AND SYSTEM BOUNDARIES

A. FROM DOCUMENTS TO A LIMITED EVIDENCE BUDGET

Let q denote a query and D a collection of parent documents. Preprocessing splits each document into one or more chunks. Dense retrieval scores a parent using its best-matching chunk, which also supplies the representative passage for relevance inference. The distinction matters because judgments apply to

parent documents, whereas the model sees bounded passages. A relevant parent does not make every one of its chunks relevant.

Write $\mathcal{C}_{64}(q)$ for the dense top-64 parents, $\mathcal{C}_{15}(q)$ for the frozen hybrid shortlist, and S for a selected set of $k = 8$ distinct parents. A ranking $\pi(S)$ orders these parents for evaluation. The evidence budget constrains $|S|$ rather than the number of tokens in a generated answer. No answer generator is invoked by the reported retrieval benchmark.

For each candidate i, the classical feature pipeline supplies query cosine c_i, lexical score b_i, a normalized representative vector v_i, and cross-encoder logit z_i. Define

$$d_{ij} = \min\{1, \max\{0, v_i^\top v_j\}\}, \qquad i \neq j. \tag{1}$$

The set redundancy measure is

$$\mathcal{R}(S) = \frac{1}{\binom{k}{2}} \sum_{\{i,j\} \subset S} d_{ij}. \tag{2}$$

This measure lies between zero and one and represents mean clipped cosine, not the percentage of duplicate documents. Negative cosine values are clipped to zero; a zero penalty does not establish that two passages are semantically independent.

The base selector maximizes a relevance–redundancy utility. The refinement instead minimizes redundancy subject to a protected prefix and neural-score conditions. These are different optimization questions. Neither is directly an optimization of the unknown test judgments. Judgments are used only after rankings have been saved, including in the journal's retrospective diagnostics.

B. CANDIDATE OPPORTUNITY AND A RECALL CEILING

For a query with nonempty set J_q of positively judged relevant parents and any candidate pool C, define judged coverage and a cardinality-aware recall ceiling as

$$\mathrm{Cov}(C, q) = \frac{|C \cap J_q|}{|J_q|}, \tag{3}$$

$$\mathrm{Ceil}_8(C, q) = \frac{\min\{8, |C \cap J_q|\}}{|J_q|}. \tag{4}$$

Every eight-document output drawn from C has Recall@8 no greater than $\mathrm{Ceil}_8(C, q)$. This follows by counting: the output can contain neither more than eight relevant documents nor more relevant documents than the pool contains. Equality is attainable for this diagnostic objective by selecting as many judged-relevant pool members as the budget permits. It is not a claim that a real system can identify them without judgments.

When one pool contains another, both diagnostic quantities are nondecreasing. Candidate expansion can therefore increase the opportunity for judged recall, but it need not increase the recall of a particular selector. Conversely, no optimizer operating only on a fixed pool can recover a relevant parent absent from that pool. This separates three effects that aggregate effectiveness often conflates: corpus-to-pool loss, pool-to-selected-set loss, and ordering within the selected set.

The diagnostic applies to observed judgments. Incompletely judged collections may contain useful unjudged documents, so the resulting ceiling bounds the official judged metric rather than every possible definition of relevance. It also ignores the refinement's prefix and score constraints. An expanded-pool ceiling is thus an optimistic relaxation of the constrained task. The search algorithm alone cannot account for a gap to this ceiling.

C. IMPLEMENTATION AND MEASUREMENT BOUNDARIES

Figure 1 distinguishes retrieval, neural relevance inference, base selection, and optional refinement. The application code also supports document ingestion and question-answering interfaces. Their existence does not imply that interface quality, optical character recognition accuracy, or live generated answers were evaluated by these experiments. The measured research pipeline uses public parent-document judgments and frozen passage construction.

The study uses two simulation models. The main selector stores amplitudes for subsets of the required size and constructs their objective values classically. The native study builds reversible circuits with explicit decision and workspace registers. It supplies bounded synthesis and validation evidence; the main selector supplies the retrieval rankings. Neither model includes device routing, noisy execution, or a measured quantum processor.

IV. RELEVANCE-GUIDED SUBSET SEARCH

A. HYBRID CANDIDATE CONSTRUCTION

All subsequent hybrid candidate calculations are restricted to the dense top-64 pool. Within this pool, let $c_{\max} = \max_i c_i$ and $b_{\max} = \max_i b_i$. The reproduced hybrid score is

$$h_i = (1 - \beta)\frac{1 + c_i}{1 + c_{\max}} + \beta\frac{b_i}{b_{\max}}. \tag{5}$$

When lexical scores are negligible, only the dense component is used. If its denominator is negligible, the dense component becomes $(1+c_i)/2$. The numerical tolerance is 10^{-12}. Hybrid score and then document ID determine the top-fifteen shortlist. This is score fusion over an already retrieved pool, not a quantum search over the document collection.

Raw cross-encoder logits remain separate from hybrid scores and dense cosine. Within the original fifteen, normalized relevance is

$$r_i = \frac{z_i - z_{\min}}{z_{\max} - z_{\min}}, \tag{6}$$

with $r_i = 0.5$ when the range is negligible. Negative logits are allowed. This normalization affects the optimization coefficients, while final relevance ordering uses the original logits. Its scale is local to a shortlist; comparing a normalized value between different pools as if it were a calibrated probability would be unwarranted.

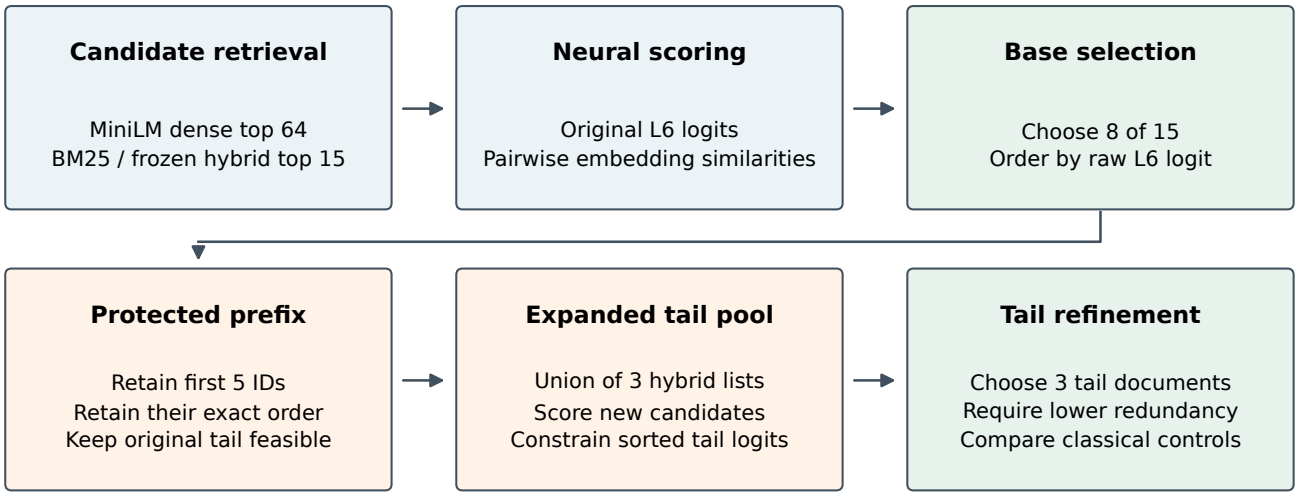


FIGURE 1. System boundaries and reported configurations. Classical retrieval and neural inference precede ideal amplitude amplification simulated on a CPU. The separate native resource study does not replace those classical stages or constitute a full hardware execution of the retrieval pipeline.

B. BASE OBJECTIVE AND INTEGER ENCODING

For $|S| = k = 8$, the base utility is

$$U(S) = \frac{1}{k}\sum_{i\in S} r_i - \lambda \mathcal{R}(S). \tag{7}$$

Both terms use classical features. With $A_b = 2^8 - 1 = 255$, define rounded coefficients

$$a_i = \mathrm{round}(A_b r_i), \tag{8}$$

$$a_{ij} = \mathrm{round}\left(\frac{2A_b\lambda}{k-1} d_{ij}\right), \tag{9}$$

$$E_b(S) = -\sum_{i\in S} a_i + \sum_{\{i,j\}\subset S} a_{ij}. \tag{10}$$

The implementation uses ties-to-even rounding. Lower integer energy is preferred. The raw-L6 top eight initialize the incumbent, and selected outputs are ordered by descending raw logit with document-ID tie resolution.

Without rounding, $-E_b(S)/(A_b k)$ equals (7). There are k unary coefficients and $\binom{k}{2}$ pair coefficients, each rounded with error at most one half. Consequently,

$$\left|\frac{-E_b(S)}{A_b k} - U(S)\right| \le \frac{k+1}{4A_b}. \tag{11}$$

This is a coefficient-approximation statement for a fixed set. It does not bound the relevance model's error or the difference between utility and judged retrieval effectiveness. At $\lambda = 0$, the raw-relevance incumbent already maximizes the relevance-only utility; the implementation retains it rather than using equal quantized energies to justify arbitrary changes in membership.

C. PROTECTED-PREFIX REFINEMENT

Let the seed-specific base ranking be $(s_1, \ldots, s_8)$, its ordered head be $H = (s_1, \ldots, s_5)$, and its original tail be $T_0 = \{s_6, s_7, s_8\}$. Construct an expanded candidate union

$$\mathcal{C}_\cup = \bigcup_{\beta\in\{0,0.25,0.5\}} \mathrm{Top}_{15}(h_\beta; \mathcal{C}_{64}). \tag{12}$$

The union contains the original shortlist and has at most 45 parents. Original-shortlist logits are retained exactly; additional inference scores only newly introduced candidates. This avoids redefining the baseline through an incidental change in inference batching.

The refinement searches $T \subseteq \mathcal{C}_\cup \setminus H$ with $|T| = 3$. Sort the tail logits in ascending order, denoted $z_{(1)}(T) \le z_{(2)}(T) \le z_{(3)}(T)$. Admissibility requires

$$z_{(j)}(T) \ge z_{(j)}(T_0) - \varepsilon, \quad j \in \{1,2,3\}, \quad \varepsilon = 10^{-12}. \tag{13}$$

The rule is a positionwise condition on sorted score values, not a requirement to retain each original tail document. Candidates below the smallest permissible score are removed before the subset basis is constructed. If M candidates survive, the cardinality basis has $\binom{M}{3}$ states, some of which can still violate the remaining score conditions.

For fixed thresholds $\theta_j = z_{(j)}(T_0) - \varepsilon$, condition (13) is equivalently expressed by

$$\sum_{i \in T} \mathbf{1}\{z_i \geq \theta_j\} \geq 4 - j, \qquad j = 1, 2, 3. \tag{14}$$

To see this, an ascending tail has its jth value at or above θ_j exactly when at least its largest $4 - j$ values meet that threshold. Inclusive comparisons handle repeated logits. The count form is useful for reversible implementation, while the sorted form is convenient for interpreting admissibility.

The refinement minimizes $\mathcal{R}(H \cup T)$. With $A_r = 2^{16} - 1$, its integer objective is

$$E_r(H \cup T) = \sum_{\{i,j\} \subset H \cup T} \mathrm{round}(A_r d_{ij}). \tag{15}$$

The full measure includes head–head, head–tail, and tail–tail terms. Head–head terms are constant during refinement but remain included in the reported redundancy. The oracle marks admissible states whose integer energy is strictly below the incumbent energy. A measured update is accepted only if its true, unrounded redundancy improves by more than ε.

The output is the original ordered head followed by the selected tail sorted by raw L6 logit. This intentionally prioritizes prefix preservation over a global re-sort: an additional candidate with a high logit does not displace a protected head document. The separate global union-L6 control evaluates a different ordering rule.

D. WHAT THE CONSTRAINTS GUARANTEE

Proposition 1 (Protected ranking metric): For fixed query judgments, preserving the first five document IDs and their order preserves nDCG@5 exactly, independently of how the last three outputs are chosen.

Proof: The numerator of nDCG@5 contains only the gains and discounts at those five ranks. Its ideal denominator is a function of the fixed judgments, not the retrieved tail. Both quantities are unchanged. The equality therefore also holds after seed and query averaging. ∎

Proposition 2 (Incumbent safety): The original tail is admissible and survives the score prefilter. Every accepted refinement retains eight unique outputs and never increases the reported redundancy relative to its starting base set.

Proof: Each sorted original-tail score meets its own bound in (13), including ties within the tolerance. Its documents therefore survive the smallest-threshold prefilter and supply a cardinality-feasible initial state. The controller replaces that state only with a valid measured tail of size three disjoint from the protected head and with strictly lower true redundancy. Induction over accepted updates proves the claim. If the search ends without improvement, its feasible incumbent is still returned. ∎

These properties do not guarantee a strict reduction for every query. A trivial remaining basis, an optimal incumbent, or a failed sampled improvement may leave the original set unchanged. Nor do they guarantee recall. A relevant original-tail document can be replaced by an irrelevant document that receives a higher cross-encoder score and has lower pair similarity. Such a replacement satisfies the neural-score constraints while reducing the number of judged-relevant documents returned. Section VII shows that this failure mode is present in the saved results.

For general pair precision b, the refinement scale is $A = 2^b - 1$. Rounding each of the 28 pair coefficients yields

$$\left| \frac{E_r(S)}{28A} - \mathcal{R}(S) \right| \leq \frac{1}{2A}. \tag{16}$$

For exact integer minimizer S_A and exact float minimizer $S_\star$ over the same admissible domain, applying this inequality twice gives

$$\mathcal{R}(S_A) - \mathcal{R}(S_\star) \leq \frac{1}{A}. \tag{17}$$

The distinction between the exact encoded optimum and the budgeted GAS output is indispensable. Equation (17) does not bound the latter's search error. At sixteen bits the per-set approximation bound is about 7.63×10^{-6}; increasing precision cannot correct a misjudged relevance score.

E. IDEAL AMPLITUDE AMPLIFICATION AND STOPPING RULES

The base cardinality basis contains $F_b = \binom{15}{8} = 6{,}435$ states. Refinement uses $F_r = \binom{M}{3}$ states. Write $|u\rangle = F^{-1/2} \sum_x |x\rangle$ for the uniform state on the corresponding cardinality basis. An oracle and reflection step are

$$O_\tau |x\rangle = (-1)^{[x \text{ admissible} \wedge E(x) < \tau]} |x\rangle, \tag{18}$$

$$G_\tau = (2|u\rangle\langle u| - I) O_\tau. \tag{19}$$

The compact simulator performs the sign changes and the reflection about the mean explicitly. Measurement draws a state according to squared amplitudes. These are ideal quantum-search operations represented and executed on a classical CPU. The controller does not use a precomputed optimum or a marked-state count to choose an ideal rotation number.

Each stage begins with its feasible incumbent and sampling window $m = 1$. A trial draws an integer from zero through $\lceil m \rceil - 1$, respecting the remaining phase budget, prepares the uniform state, applies the sampled number of steps, and measures. A strict accepted improvement resets the window; otherwise the window grows by a factor of 1.2, capped at $\sqrt{F}$. Recorded limits are 64 phase calls, 32 trials, and a cooperative 15-second time limit for each stage and seed. Base and refinement have separate budgets, so a refined request may use up to 128 phase calls.

Table 2 exposes an important implementation cost: the compact simulator evaluates the complete basis before amplification. The controller's phase-call allowance is therefore only one accounting unit. Exact enumeration can inspect the same prepared objective array without amplitude simulation. This prevents a low logical phase count from being interpreted as evidence of lower wall-clock work for the implemented selector.

TABLE 2. Refinement controller in implementation order. Integer marking and true-redundancy acceptance have different roles; neither uses test judgments.

Require: Original ranked eight B, expanded scored pool C, seed, and recorded stage budgets
Set $H \leftarrow B[1{:}5]$; $T_0 \leftarrow B[6{:}8]$; incumbent $T \leftarrow T_0$; window $m \leftarrow 1$
Preserve original L6 logits; build inclusive score thresholds from T_0
Remove tail candidates below the smallest threshold; enumerate the size-three compact basis
Construct integer energies and admissibility flags classically for that basis
while phase, trial, and cooperative time limits permit another trial **do**
  Draw a rotation count from the current window, restricted by the remaining phase allowance
  Prepare $|u\rangle$; phase-mark admissible integer improvements; reflect; measure a candidate tail T'
  if T' is admissible, has lower integer energy, and $\mathcal{R}(H \cup T') < \mathcal{R}(H \cup T) - \varepsilon$ **then**
    Accept $T \leftarrow T'$ and reset $m \leftarrow 1$
  else
    Grow $m \leftarrow \min\{1.2m, \sqrt{F}\}$
  end if
end while
return Original ordered H, followed by T sorted by raw L6 score; retain counts and termination status

V. DATA AND EXPERIMENTAL PROTOCOL

A. COLLECTIONS, SPLITS, AND MODEL IDENTITIES

Table 3 lists the three collections, their complete judged test-query counts, and the frozen base settings. All positive-query IDs in each official test split are evaluated. The reported evidence comprises 300 SciFact, 323 NFCorpus, and 648 FiQA queries; no partially collected fourth collection is included. The separate resource study uses saved features from these same domains; it is not another retrieval-effectiveness test.

The dense encoder is sentence-transformers/all-MiniLM-L6-v2, producing 384-dimensional normalized vectors. The main relevance model is cross-encoder/ms-marco-MiniLM-L6-v2; an L2 cross-encoder supplies a smaller neural control. Full repository revisions and model-file hashes are retained in the evidence manifest. These identities distinguish a named model family from the exact artifacts used for a run. The journal does not download new model weights or rerun inference.

Text normalization uses Unicode NFKC, whitespace collapse, and deterministic handling of document text. Tokenizer offsets are computed without padding or truncation. A chunk contains at most 190 WordPiece content tokens, advances by 158 content tokens, and includes two special tokens in a model input of at most 192 tokens. Every parent score is the maximum query cosine over its chunks; the lowest chunk index resolves an exact maximum tie. The selected original chunk is the representative passage. This policy preserves a common parent ID and passage representation across compared methods sharing a pool.

Queries are checked against the content-token limit, and cross-encoder query–passage pairs are checked against the 512-token limit. Silent truncation is not an accepted substitute for these checks. Preserving the original fifteen logits during expansion further prevents a changed neural batching context from perturbing the baseline result. Corpus, query, and judgment hashes bind the saved inputs to the recorded experiments.

B. DEVELOPMENT DECISIONS AND FOLLOW-UP STATUS

The fixed SciFact development sample contains one fifth of the official training queries. Tuning on NFCorpus and FiQA uses the supplied development splits. The base parameter search covers

$$\begin{aligned} &\beta \in \{0, 0.25, 0.5\}, \\ &\lambda \in \{0, 0.01, 0.025, 0.05\}. \end{aligned} \tag{20}$$

All stochastic settings use seeds zero through four. A development setting is eligible if its mean nDCG@5 and Recall@8 match or exceed L6-only on the same shortlist within 10^{-12}. Eligible configurations are ordered lexicographically: maximize nDCG@5 first and Recall@8 second; then minimize redundancy, followed by λ and finally β.

This selection rule permits zero diversity weight. SciFact's selected zero weight is informative: under this configuration the base's relevance-ranked incumbent is already optimal for its unquantized relevance objective. It would be incorrect to infer a nonzero diversity effect merely because the search mechanism is called GAS. The objective, its frozen weights, and the search process must be reported separately.

The L6 MMR tuning grid contains six values: 0, 0.1, 0.25, 0.5, 0.75, and 1. The recorded relevance-first preference selects zero diversity weight on these collections. Its measured row therefore represents one selected operating point, not the full MMR tradeoff frontier. Cosine-guided controls retain their own frozen settings. Legacy threshold selection considers $\{0.3, 0.5, 0.7\}$ on development nDCG@5, with the declared tie preferences.

The refinement confirmation protocol uses deterministic 100-query training samples, excluding historical development, pilot, and test IDs. Its descriptive acceptance rule combines three checks against the matched base: equal mean nDCG@5, a mean-recall ratio of at least 0.98, and a strict decrease in mean redundancy. NFCorpus's confirmation lost 2.1561% relative recall and failed this gate. An explicitly recorded continuation then collected its full test without changing the method or the global gate. The later full-test

TABLE 3. Recorded datasets and configurations. Documents are parent documents. SciFact development is derived from official training; the other two use official development splits. The lexical weight is β and the base redundancy weight is λ.

Dataset	Documents	Dev.	Test	β	λ	Reported policy
SciFact	5,183	161	300	0.25	0.000	Base + refinement
NFCorpus	3,633	324	323	0.50	0.050	Base + refinement
FiQA	57,638	500	648	0.00	0.025	Base

outcome does not retroactively turn that failed confirmation into a pass.

Earlier test outcomes had already been observed during method development. The newer official tests must consequently be described as follow-up evaluations. The reported parameters were not retuned against those follow-up judgments, but the tests are not an untouched holdout for the redesign. The journal diagnostics are retrospective descriptions of these same saved runs.

C. CONTROLS AND COMPARISON FAIRNESS

The original evaluation contains twenty methods; protected-refinement runs add seven expanded-pool policies. They include dense cosine, dense/BM25 RRF with rank constant 60, native cosine and L6 MMR, cosine-guided GAS, L6-only ranking, base L6-guided GAS, greedy and exact optimizers, random search, legacy threshold Grover, and L2/L6 neural controls. Cross-encoder controls on dense fifteen and dense sixty-four are distinct from L6-only on hybrid fifteen.

The seven expanded-pool policies are constrained GAS, constrained local-swap greedy, exact float, exact quantized, matched random search, global union-L6 ranking, and union-L6 with the protected prefix. Constrained policies share each seed's actual baseline prefix and the same tail feasibility rules. The random refiner draws $\max(1, 2Q)$ uniform subsets plus its initial incumbent, where Q is that seed's recorded refinement phase count. This matches a specified objective-access allowance; it does not match every operation performed by the compact simulator.

Exact controls independently enumerate their own float or quantized objectives. The GAS controller receives neither their optimum nor the count of marked states. A comparison across different candidate pools characterizes complete policies. A comparison among constrained refiners on identical features is more specific: it isolates their search outcomes for the same objective and admissible domain. These two kinds of comparison support different causal interpretations.

D. METRICS, INFERENCE, AND HARDWARE

The principal metrics are nDCG@5, Recall@8, and redundancy from (2). We also report nDCG@8 to expose effects at ranks six through eight. Discounted gains use linear judgment values and logarithmic rank discount [34]; saved-ranking checks use NIST's trec_eval [35]. A five-seed policy is averaged within each query before queries are averaged. Five seeds do not provide five independent relevance judgments for the same query.

The recorded protected-run inference uses 10,000 paired bootstrap resamples with seed zero and 10,000 paired sign-randomization draws with seed one. Numerical differences no larger than 10^{-12} are ties. Holm correction covers nine comparators by four metrics for each protected dataset. The earlier FiQA nDCG audit covers its six declared comparators and removes numerical averaging artifacts before testing. We retain those families and pointwise intervals; no new selected family is substituted to improve apparent significance.

The recorded host is an Intel Core i7-1355U with 12 logical CPUs and approximately 15.3 GiB of RAM, running Linux and Python 3.12.3. Compute libraries use two threads. The recorded software includes NumPy 1.26.4, SciPy 1.13.1, CPU PyTorch 2.12.0, Transformers 4.57.6, Sentence Transformers 3.4.1, Qiskit 0.46.0, and Aer 0.13.3 [52]. Model loading, corpus/cache restoration, and artificial-query warm-up are outside online latency. Session and host-load records remain available; the experiments were not a dedicated latency laboratory.

Online time composes the applicable retrieval, lexical, neural, and selection stages. Reused upstream computation is attributed to every policy that requires it. Combined selector percentiles use the actual sum of base and refinement selection time for each seeded request before aggregation. Request P95 is not divided by the number of seeds. Section VII adds an additive mean-cost decomposition and explicitly avoids interpreting sums of component percentiles as pipeline percentiles.

VI. RECORDED RETRIEVAL RESULTS

A. EFFECTIVENESS UNDER THE DECLARED CONFIGURATIONS

Table 4 reports the selected full-test policies. The values come from the completed studies; no new journal experiments were collected. The reported nDCG@5 values are 0.6878, 0.3689, and 0.3575 on SciFact, NFCorpus, and FiQA, respectively. In the first two rows, nDCG@5 is tied to the matched base by prefix preservation. FiQA's row instead reports the earlier fifteen-candidate base selector. The configuration label is therefore necessary to interpret the result.

Relative to matched base selection, SciFact refinement reduces mean redundancy by 3.21% and changes mean Recall@8 by -0.93% relative. NFCorpus reduces redundancy by 2.63% with a +1.95% relative recall change. The recall changes have different signs across domains. The refinement therefore cannot be described as a general recall improvement, even though the same prefix metric is preserved on both

TABLE 4. Reported full-test policies. SciFact/NFCorpus include refinement; FiQA uses the earlier base. All quantities within a row come from the same recorded policy.

Dataset	nDCG@5	nDCG@8	Recall@8	Redundancy ↓
SciFact	0.687833	0.698173	0.801111	0.433636
NFCorpus	0.368865	0.346506	0.152997	0.431771
FiQA	0.357490	0.376939	0.434063	0.531244

collections.

The FiQA base reaches Recall@8 of 0.4341 and redundancy of 0.5312. It matches the reported nDCG@5 and recall of its same-shortlist L6-only control. A later, separately completed 648-query FiQA refinement follow-up preserved nDCG@5 and reduced redundancy by 8.23%, but lost 2.60% relative recall. That result failed the frozen 98% recall-retention gate and was not retuned into a passing result. We retain the prior manuscript's disclosure of this failure without substituting its redundancy value into the earlier base row.

B. CLASSICAL AND NEURAL COMPARISONS

Table 5 places the reported policy alongside the declared controls. The cross-encoder shortlist distinctions matter: L6-only uses hybrid fifteen, while the dense-fifteen neural control uses the fifteen highest dense-scoring parents. Differences between those rows include candidate construction. Dense-sixty-four neural reranking additionally exposes the relevance model to more candidates and incurs the corresponding inference cost.

Matched greedy and exact-float refiners have the same aggregate nDCG@5, nDCG@8, Recall@8, and redundancy as constrained GAS on the SciFact and NFCorpus tests. This agreement shows that the measured quality of those outputs does not depend on the Grover search mechanism. The constrained objective and the available candidates explain a benefit that several optimizers can realize.

Table 6 further separates candidate expansion, prefix retention, and constrained set optimization. Global union-L6 permits complete reordering. Protected-prefix union-L6 preserves the head but chooses the tail by neural relevance. The constrained optimizers additionally optimize pair redundancy under the score thresholds. These controls address different sources of change; grouping them under one generic ''reranking'' label would obscure the mechanism being evaluated.

The observed tradeoffs are visualized in Fig. 2. Some inexpensive or less relevance-focused policies have lower redundancy while also having lower recall. The graph contains measured operating points, not a tuned Pareto frontier or a proof of dominance over every possible setting. In particular, the zero-diversity MMR point does not represent the complete range of MMR operating choices.

C. PAIRED UNCERTAINTY AND INTERPRETATION

Table 7 reproduces paired differences against the base for the protected datasets. SciFact's recall interval includes both a loss and a gain, while its mean is negative. NFCorpus's positive recall interval is pointwise; the recorded raw probability does not produce a rejection after the complete Holm family is considered. The distinction between a displayed interval and a familywise testing decision must remain explicit.

The empirical 98% mean-recall gate is not a statistical noninferiority test, and an interval containing zero does not establish equivalence. Stronger inferential claims would require a justified noninferiority margin, a prospective analysis, and data whose role in method development is appropriately controlled. The present evidence supports the recorded descriptive gate outcomes and the stated paired comparisons.

The FiQA statistical audit also illustrates why floating-point conventions matter. Averaging five identical seeded scores can produce a difference of approximately one unit in the last place relative to a deterministic control. A significance test must not promote that numerical artifact into evidence of a retrieval improvement. The recorded 10^{-12} tie policy addresses this problem while leaving rankings and reported raw means intact.

D. LATENCY AND OPTIMIZATION OUTCOMES

Complete online median and P95 times appear in Table 8. Expanded-pool inference increases complete latency relative to the matched base. A small refinement time does not make the two-stage pipeline equally inexpensive. Comparisons with dense-sixty-four rerankers involve both cost and candidate opportunity and should not be reduced to a universal speed claim.

Table 9 distinguishes logical work from outcome quality. An exact objective hit is evaluated against the corresponding exact float-redundancy optimum for refinement and the exact integer optimum for the base. Such a hit certifies neither maximum judged recall nor maximum nDCG. Budget exhaustion, a trivial basis, and an unchanged incumbent are meaningful outcomes retained in the request denominator.

The work counts document this implementation's behavior. They cannot establish physical acceleration: the CPU simulator prepares the complete objective array, and classical methods can inspect that array directly. The objective assumptions and measured cost boundaries remain explicit.

VII. JOURNAL EXTENSION: COVERAGE, FAILURE, AND COST DIAGNOSTICS

A. CANDIDATE COVERAGE SEPARATES RETRIEVAL FROM SELECTION

We reconstruct candidate membership from the saved feature and ranking records. It evaluates (4) on dense sixty-four, hybrid fifteen, and, where available, the expanded union. This analysis produces no new rankings. Before calculating coverage, we independently check pool sizes, subset inclusion, and parent IDs so that historical field names do not obscure the actual pools.

Table 10 and Fig. 3 show that candidate opportunity is substantially collection-dependent. SciFact's hybrid-fifteen ceiling is 0.8569, and its expanded-union ceiling is 0.8903.

TABLE 5. Recorded full-test comparison. n@5 is nDCG@5, R@8 is Recall@8, and Red. is redundancy. Matched search controls use the refined domain on SciFact/NFCorpus and the base domain on FiQA. Additional saved methods remain in the complete CSV export.

	SciFact			NFCorpus			FiQA		
Policy	n@5	R@8	Red.	n@5	R@8	Red.	n@5	R@8	Red.
Dense cosine	0.6361	0.8111	0.4756	0.3181	0.1335	0.4398	0.3470	0.4153	0.5363
Dense/BM25 RRF	0.6677	0.8149	0.4727	0.3491	0.1394	0.4394	0.3509	0.4262	0.5327
Legacy threshold Grover	0.6364	0.8111	0.4756	0.3181	0.1334	0.4398	0.3458	0.4150	0.5363
Cosine-guided GAS	0.6311	0.7742	0.4490	0.3160	0.1235	0.3996	0.3452	0.4093	0.5215
L6 relevance only	0.6878	0.8086	0.4480	0.3689	0.1500	0.4447	0.3575	0.4341	0.5313
L6 MMR	0.6878	0.8086	0.4480	0.3689	0.1500	0.4447	0.3575	0.4341	0.5313
Base L6-guided GAS	0.6878	0.8086	0.4480	0.3689	0.1501	0.4434	0.3575	0.4341	0.5312
Matched greedy	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5311
Matched exact float	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4343	0.5309
Matched exact integer	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5311
Matched random	0.6878	0.8011	0.4338	0.3689	0.1531	0.4320	0.3575	0.4341	0.5313
L2, dense 15	0.6431	0.8049	0.4642	0.3509	0.1396	0.4486	0.3319	0.4217	0.5295
L6, dense 15	0.6790	0.8104	0.4655	0.3621	0.1423	0.4537	0.3575	0.4341	0.5313
L2, dense 64	0.6321	0.7916	0.4329	0.3481	0.1438	0.4468	0.2973	0.3918	0.4962
L6, dense 64	0.6815	0.8087	0.4379	0.3670	0.1437	0.4586	0.3535	0.4160	0.5033
Reported GAS policy	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5312

TABLE 6. Expanded-pool controls on the two protected-refinement test runs. All four metrics are shown to expose tail effects hidden by a fixed top five. FiQA has no corresponding expanded-pool headline in the selected source mapping.

	SciFact				NFCorpus			
Policy	n@5	n@8	R@8	Red.	n@5	n@8	R@8	Red.
Union L6, global order	0.6860	0.6972	0.8037	0.4475	0.3688	0.3472	0.1522	0.4477
Union L6, protected prefix	0.6878	0.6993	0.8037	0.4474	0.3689	0.3466	0.1523	0.4474
Constrained GAS	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained greedy	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained exact float	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained exact integer	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained random	0.6878	0.6982	0.8011	0.4338	0.3689	0.3468	0.1531	0.4320

TABLE 7. Recorded paired inference: reported refinement minus base on SciFact/NFCorpus. †FiQA is its earlier base minus L6-only for the independently audited nDCG@5 comparison. Intervals are pointwise, not simultaneous; Holm decisions use each original full family.

Dataset	Metric	Mean difference	Paired 95% CI	Raw p	Holm reject
SciFact	nDCG@5	+0.00000	[+0.00000, +0.00000]	1.00000	No
SciFact	nDCG@8	-0.00268	[-0.00811, +0.00231]	0.30427	No
SciFact	Recall@8	-0.00750	[-0.02417, +0.00750]	0.45645	No
SciFact	Redundancy	-0.01439	[-0.01680, -0.01216]	0.00010	Yes
NFCorpus	nDCG@5	+0.00000	[+0.00000, +0.00000]	1.00000	No
NFCorpus	nDCG@8	+0.00094	[-0.00248, +0.00445]	0.58974	No
NFCorpus	Recall@8	+0.00293	[+0.00038, +0.00622]	0.03690	No
NFCorpus	Redundancy	-0.01165	[-0.01405, -0.00938]	0.00010	Yes
FiQA	nDCG@5†	0.00000	[0.00000, 0.00000]	1.00000	No

NFCorpus's corresponding ceilings are 0.1725 and 0.1942. FiQA's base shortlist has a ceiling of 0.4824, compared with 0.6483 in dense sixty-four. These are optimistic judged opportunities, not estimates of what a learned reranker will necessarily attain.

There are 40 SciFact, 91 NFCorpus, and 203 FiQA queries with no positively judged parent in the hybrid shortlist. An eight-of-fifteen selector cannot improve judged recall for those queries without first changing the available candidates. Expansion can repair some such exclusions on the protected datasets, but it remains confined to dense sixty-four and cannot recover an omitted corpus document outside that set.

The ceiling also clarifies why raw Recall@8 is not directly comparable across datasets. If a query has many judged-relevant documents, even eight perfectly chosen outputs retrieve only a fraction of them. NFCorpus includes queries with much larger relevant sets than the single-evidence cases common in SciFact. Its lower recall scale therefore combines task structure, candidate opportunity, and ranking performance. A useful interpretation compares policies on the same

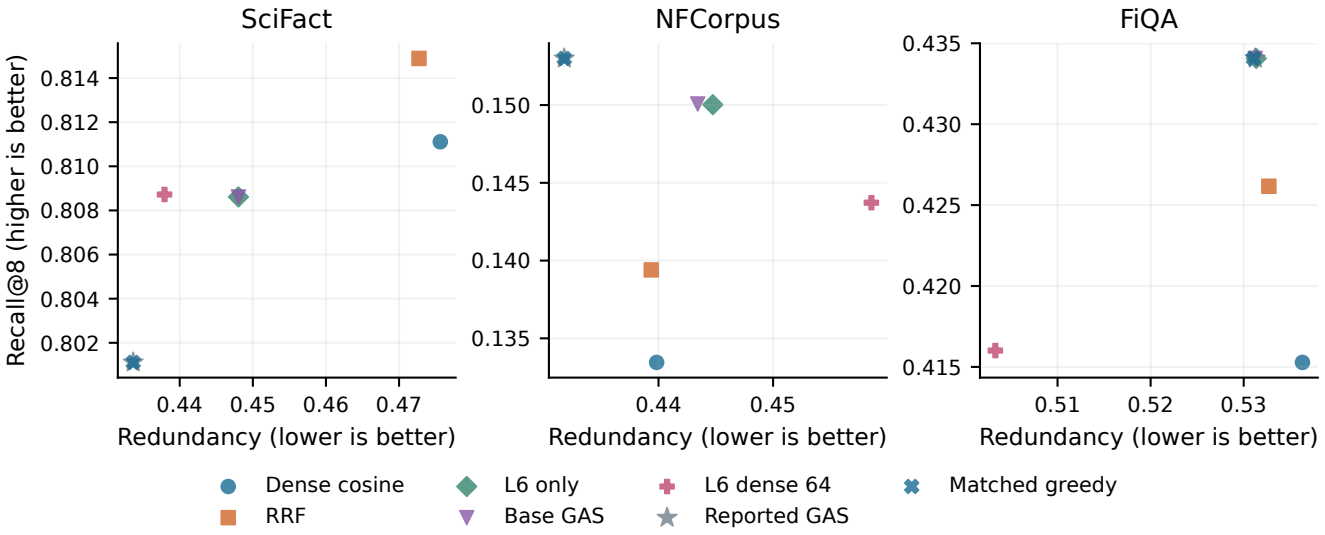


FIGURE 2. **Observed Recall@8–redundancy operating points from the saved runs. Coincident points can arise when policies return the same effective outputs. Candidate pools differ as specified in the protocol; the graph is descriptive and does not imply matched causal comparisons across every point.**

TABLE 8. **Composed online latency, milliseconds per seeded request. Values include all applicable upstream stages. P50/P95 are computed over recorded request totals; restoration, model loading, and warm-up are separate.**

	SciFact		NFCorpus		FiQA	
Policy	P50	P95	P50	P95	P50	P95
Dense cosine	33.3	48.3	24.5	32.9	79.9	98.5
Dense/BM25 RRF	61.6	94.9	31.0	44.5	225.3	328.3
Legacy threshold Grover	38.2	313.6	25.2	281.6	84.3	231.9
Cosine-guided GAS	42.3	56.8	33.7	42.0	84.1	102.5
L6 relevance only	955.3	1085.4	851.9	901.4	650.0	821.1
L6 MMR	955.5	1085.7	852.1	901.6	650.1	821.3
Base L6-guided GAS	963.2	1093.7	859.8	909.1	653.8	826.7
Matched greedy	1405.4	1670.7	1145.5	1479.6	650.0	821.2
Matched exact float	1405.2	1670.6	1145.3	1479.3	651.2	822.5
Matched exact integer	1405.2	1670.6	1145.3	1479.3	651.1	822.4
Matched random	1405.2	1670.6	1145.4	1479.3	651.9	824.8
L2, dense 15	348.0	403.0	310.5	331.5	238.9	294.6
L6, dense 15	915.8	1029.2	832.5	882.5	539.6	618.2
L2, dense 64	1374.7	1562.5	1242.5	1319.6	763.7	912.2
L6, dense 64	3799.0	4322.2	3460.8	3630.9	2037.9	2437.9
Reported GAS policy	1406.5	1672.3	1146.8	1481.3	653.8	826.7

queries and relates their outputs to the appropriate candidate ceiling.

The union enlarges the relaxed opportunity set, but the final constrained task can remain difficult for reasons unrelated to quantum search. Five output positions are committed to the original head; admissible replacements must satisfy neural-score dominance; and redundancy is only a proxy for complementary evidence. The remaining gap to the union ceiling includes those constraints and model errors. Attributing the entire gap to a finite search budget would overstate what this diagnostic establishes.

B. MEAN IMPROVEMENTS COEXIST WITH INDIVIDUAL LOSSES

Table 11 counts query-level effects after averaging seeds within each query. SciFact has 2 recall gains, 5 losses, and 293 ties relative to base GAS. NFCorpus has 15 gains, 13 losses, and 295 ties. Its favorable mean therefore coexists with harmed queries. FiQA has no recall differences beyond the tie tolerance relative to L6-only, even though some selected memberships and redundancy values differ.

Refinement lowers redundancy for 158 SciFact queries and 135 NFCorpus queries. It raises redundancy for none, as required by Proposition 2. Figure 4 shows the paired recall and redundancy changes. The cumulative distributions reveal many zero changes and the small subset of queries with recall differences, which a mean alone conceals.

TABLE 9. Recorded search work and combined selector latency in milliseconds. The first phase column counts inherited base work for refined policies; it is zero for base-only FiQA. The next column counts the current stage. Optimum-hit definitions follow the text.

Dataset	Requests	Base phases	Refine/base phases	Optimum-hit rate	Selector P50	Selector P95
SciFact	1500	64.00	20.08	1.000	9.66	10.67
NFCorpus	1615	63.91	17.82	1.000	9.49	11.02
FiQA	3240	0.00	64.00	0.994	3.95	4.54

TABLE 10. New secondary analysis of saved candidate pools. Judged coverage includes every positively judged parent in the pool. The Recall@8 ceiling additionally enforces the eight-output budget and relaxes relevance-score and prefix constraints. Values are query means.

Dataset	Pool	Mean size	Judged coverage	Recall@8 ceiling
SciFact	Dense 64	64.00	0.9243	0.9243
SciFact	Hybrid 15	15.00	0.8569	0.8569
SciFact	Expanded union	22.30	0.8903	0.8903
NFCorpus	Dense 64	64.00	0.2666	0.2375
NFCorpus	Hybrid 15	15.00	0.1780	0.1725
NFCorpus	Expanded union	20.29	0.2028	0.1942
FiQA	Dense 64	64.00	0.6489	0.6483
FiQA	Hybrid 15	15.00	0.4824	0.4824

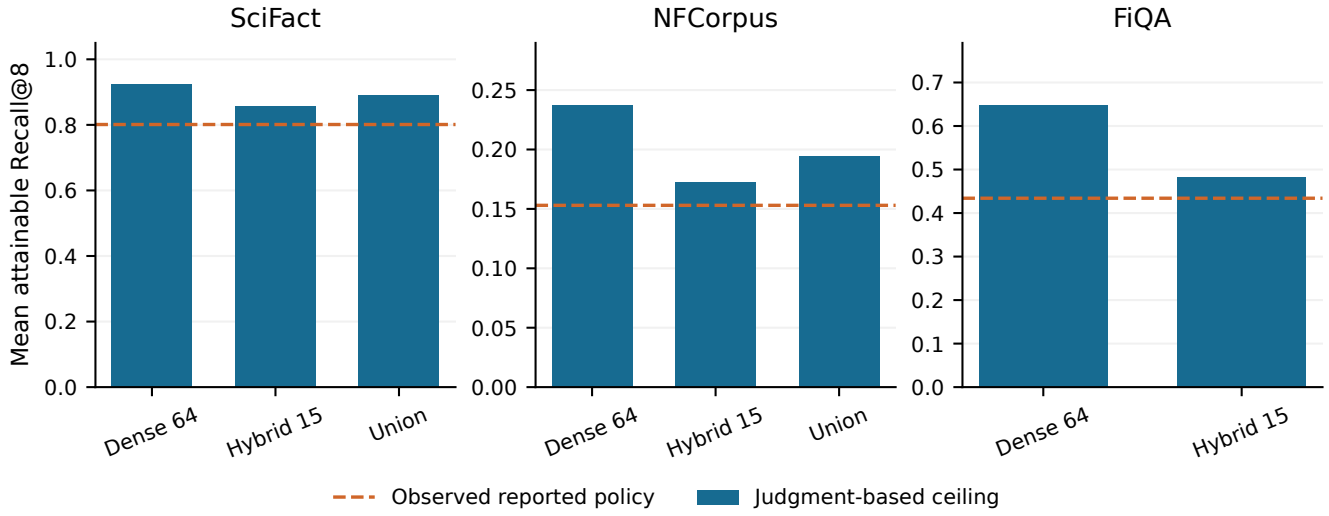


FIGURE 3. New judgment-based Recall@8 ceilings for the saved candidate pools, compared with the observed reported-policy recall. The dashed line is the same policy within each dataset panel. Expanded-union ceilings relax the protected-prefix and tail-score constraints and therefore do not measure pure search inefficiency.

Two deterministically selected extreme cases illustrate the risk without claiming representativeness. SciFact query 384 has one positively judged relevant parent. Base recall is 1, whereas refined recall is 0, despite a candidate-pool ceiling of 1 and unchanged top-five output. The sole relevant output was therefore lost below the protected head. Redundancy decreases by approximately 0.04373. The case supplies an observed instance in which the optimization proxy improves while judged retrieval becomes worse.

For NFCorpus query PLAIN-2570, the judgments contain 40 relevant parents. Base Recall@8 is 0.075 and refined Recall@8 is 0.025, corresponding to three versus one positively judged output. Mean redundancy decreases by approximately 0.09454. Both the original and expanded pool have a relaxed Recall@8 ceiling of 0.125 for this query. Additional candidate opportunity alone therefore does not explain the loss; admissibility, relevance estimates, and the selected objective also matter. These cases were chosen by the lowest paired recall change, then largest redundancy decrease, then query ID, using a fixed retrospective rule.

The practical implication is limited but important. Preserving a top-five relevance metric can be appropriate when the application explicitly values that prefix, yet it does not establish that the full evidence set is safe to replace. A mean-retention threshold can also hide a severe loss on a rare query. Judged recall distributions, the number of harmed queries, and application-specific costs of those losses should accompany the mean.

TABLE 11. New query-level outcome and seed-stability analysis. References are base GAS for SciFact/NFCorpus and L6-only for FiQA. Counts use a 10^{-12} numerical tie tolerance. Seed-varying ranks count queries with more than one final ordered list among the five saved seeds.

Dataset	Recall wins	Ties	Losses	Lower redundancy	Seed-varying ranks	Seed-varying recall
SciFact	2	293	5	158	0	0
NFCorpus	15	295	13	135	29	3
FiQA	0	648	0	6	7	0

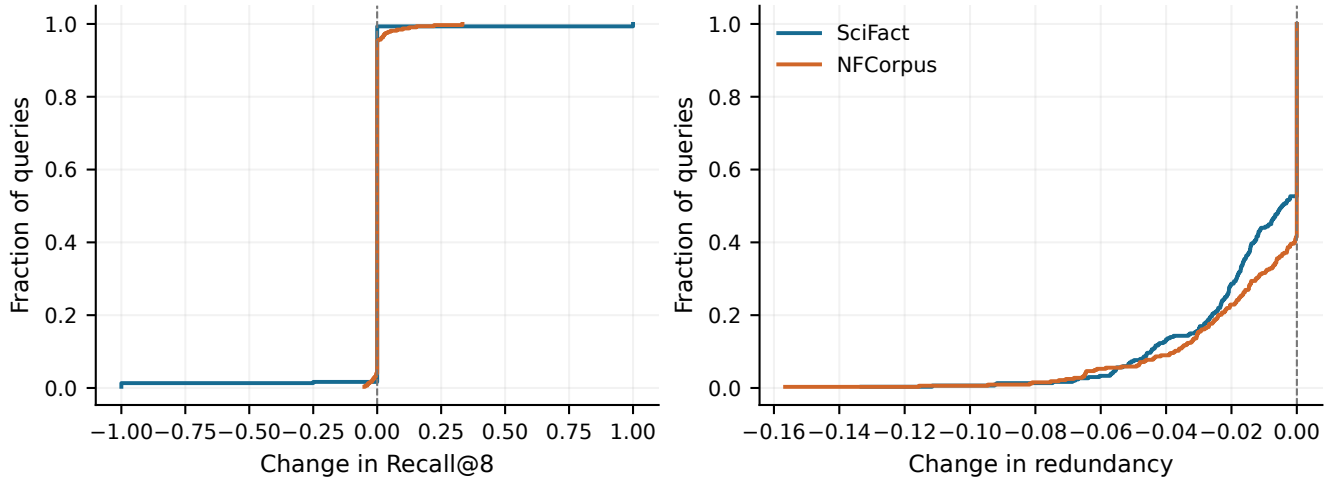


FIGURE 4. New empirical cumulative distributions of query-level refinement-minus-base differences. Recall changes can be positive or negative. Nonpositive redundancy changes follow the refinement acceptance rule. Seeds are averaged within queries before these distributions are constructed.

C. SEED STABILITY IS A PROPERTY OF OUTPUTS, NOT INDEPENDENT EVIDENCE

The final reported rankings vary across seeds for 0 SciFact queries, 29 NFCorpus queries, and 7 FiQA queries. Recall itself varies across seeds for 0, 3, and 0 queries, respectively. Deterministic final outcomes can arise even when an algorithm samples different internal trajectories. The distinction between trajectory randomness and output variability is important when interpreting a five-seed average.

Stable outcomes do not increase the number of independent queries. Conversely, seed variability in the selected tail does not necessarily change recall because different documents can have the same judged status. NFCorpus demonstrates both effects: the number of queries with varying ordered lists exceeds the number with varying recall. The complete seed export retains individual scores and search statuses, permitting this distinction to be inspected rather than inferred from aggregate variances.

These findings also limit the role of simulation in the observed quality gains. A stochastic search that consistently reaches an output also found by matched greedy and exact methods provides no evidence that randomness or amplitude amplification uniquely caused the effectiveness result. It does, however, characterize the stability of this specific implementation under its frozen budgets and seeds.

D. ADDITIVE COST ATTRIBUTION AND THE SCOPE FOR SELECTOR ACCELERATION

For each primary seeded request, the journal exporter reconstructs

$$T_{\text{online}} = T_{\text{upstream}} + T_{\text{L6,original}} + T_{\text{L6,extra}} + T_{\text{base}} + T_{\text{refine}}. \tag{21}$$

Upstream time includes query encoding, dense search, representative features, and applicable lexical/hybrid/union processing. Extra inference and refinement are zero for the FiQA base policy. The reconstructed sum is checked against the saved request total before means are calculated. Additivity applies to these composed stage measurements; it does not imply that separately measured percentile values are additive.

Figure 5 shows that original L6 inference alone accounts for 64.0%, 71.3%, and 64.8% of the respective primary mean online times. Additional L6 scoring further dominates the refined pipelines. Combined base-plus-refinement selection accounts for 0.67%, 0.88%, and 0.61%, respectively. Thus, even an idealized removal of all measured selector time would leave most of the composed request cost.

More formally, for $f = \overline{T}_{\text{selector}}/\overline{T}_{\text{online}}$ and unchanged upstream stages, replacing selection by a hypothetical zero-time implementation could reduce this mean by only fraction f. The corresponding idealized speedup is $1/(1-f)$. This is an accounting bound for the same staged computation, not a prediction of hardware performance. A practical quantum implementation would additionally require state preparation,

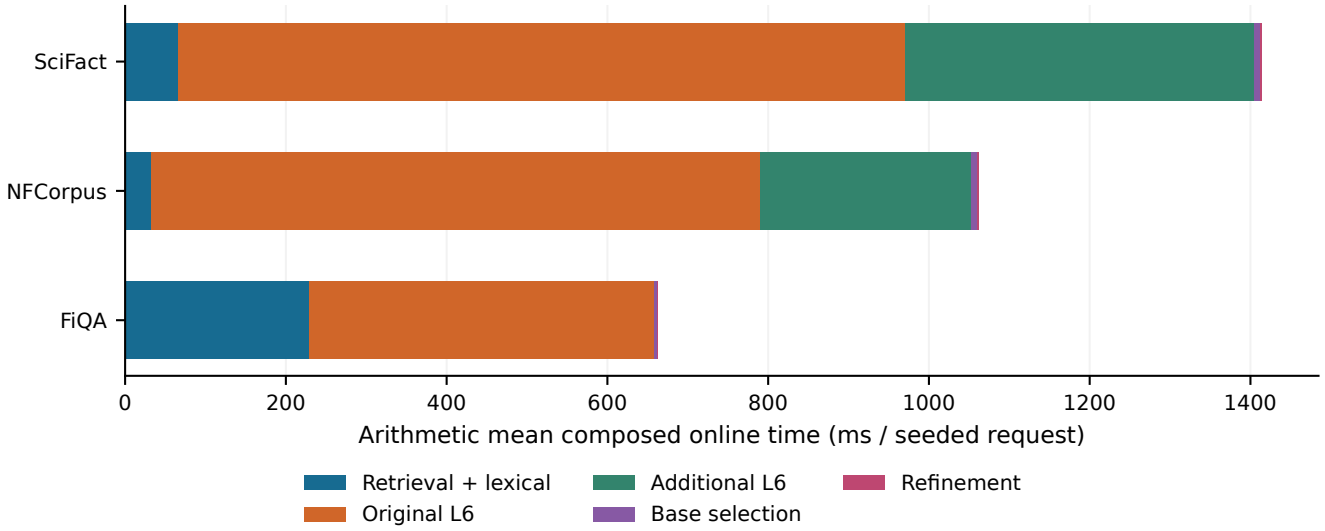


FIGURE 5. New decomposition of arithmetic mean online time using additive per-request components. Small selector segments are retained. The plot uses means for additivity; Table 8 separately reports request-level medians and P95 values. Differences between collection sessions are not controlled cross-session speed experiments.

data access, reversible arithmetic, and measurement, none of which can be assumed free. The measured cost boundary therefore gives little support to an end-to-end acceleration claim based solely on reducing selection phase counts.

VIII. RECORDED RESOURCE STUDY AND PRECISION SENSITIVITY

A. COMPACT SIMULATION AND NATIVE CIRCUITS SOLVE DIFFERENT ACCOUNTING TASKS

The compact selector represents only fixed-cardinality subsets. Its complex128 amplitude array uses $16F$ bytes, where $F = \binom{15}{8}$ for the base or $F = \binom{M}{3}$ for refinement. This allocation excludes stored coefficients, energies, candidate features, and process memory. A full ν-qubit statevector would instead require $16\,2^{\nu}$ bytes for amplitudes alone. Neither expression measures total system memory.

Compact simulation enumerates basis energies before amplification, so its objective evaluations must be counted separately from incumbent checks, measured trials, and logical phase calls. The recorded convention assigns two reversible objective calls to a phase operation for compute and uncompute. Constraint evaluation, state preparation, reflection, and physical data access would add work to an actual circuit execution. A logical query count does not capture all of these costs.

The native study uses 32 saved training queries per collection, drawn from the confirmation samples. It evaluates five compact-search seeds, tail pools of 3, 5, 8, 12, 16, or 20 candidates, and pair precisions of 4, 8, 12, or 16 bits. Its 178 completed tasks include 80 native synthesis cells. Actual pool availability is retained: a requested size unavailable for a real query is not silently replaced by a smaller pool. Artificial stress instances at the specified larger sizes are labeled separately.

The study was completed before this journal manuscript. It provides neither new relevance judgments nor another full-test evaluation. Its controlled subpools retain the original tail, and native representatives are the first two eligible frozen seed-zero queries for a dataset/pool, when available. The representatives remain fixed across precisions. Comparisons across pool sizes may nevertheless involve different eligible queries, so growth curves describe the measured cells rather than a random population scaling law.

B. REVERSIBLE CONSTRAINT AND OBJECTIVE REGISTERS

Writing $x_i \in \{0, 1\}$ for a remaining tail candidate, the integer refinement objective has the polynomial form

$$E(x) = C_H + \sum_i \alpha_i x_i + \sum_{i<j} \alpha_{ij} x_i x_j, \tag{22}$$

where C_H contains fixed head–head terms, α_i sums rounded head–candidate terms, and α_{ij} represents a rounded candidate pair. The oracle evaluates signed improvement relative to the incumbent and marks only states with cardinality three and all three threshold-count conditions in (14).

The register allocation comprises M decision qubits, w signed-value qubits, $c = \lceil \log_2(M+1) \rceil$ counter qubits, $c-1$ comparator workspace qubits, and four constraint flags. Thus,

$$\nu = M + w + 2c + 3. \tag{23}$$

The signed width must bound the objective difference on all bit strings, including states with invalid cardinality. Designing w from valid three-element subsets alone could permit overflow on states the circuit must process before rejecting them. The implementation bounds the full polynomial domain and includes every workspace register in the reported total.

The oracle computes objective and constraints, conditionally applies the phase, and reverses the computation. The

TABLE 12. Recorded native-oracle synthesis summary. Slash-separated entries compare 4/16-bit coefficients; full four-precision measurements are retained in the accompanying native CSV. Width includes workspace. Depth and CX values are maxima over the stated bounded sample; stress rows are artificial.

Tail	Source	Qubits	Depth	CX gates
3	real	19/31	1049/1865	992/2408
5	real	23/35	3891/5955	3256/6160
8	real	29/41	15067/20722	11684/18240
12	real	34/46	27570/41294	22068/36076
16	stress	39/51	61665/67687	43628/52212
20	stress	43/55	79769/96137	58220/75556

same counter and comparison workspace can be reused for successive threshold checks, while their flags are retained until phase marking. This structure explains both the logical-width formula and why a compact classical basis dimension is not an interchangeable qubit count.

Table 12 summarizes the measured cells. Figure 6 includes all four precision levels and distinguishes real samples from artificial stress cases. Synthesis uses the unrestricted u, cx basis, optimization level one, and transpiler seed zero. These are logical resources without coupling-map routing, device calibration, or error correction. The gate count therefore cannot be described as the cost of execution on a particular processor.

C. PRECISION AND SEARCH ERROR

The coefficient bound in (16) decreases with b. The observed GAS output can nevertheless have a larger gap to the exact float optimum because a finite search budget need not find the exact integer optimum. Figure 7 separates this empirical search outcome from a statement about coefficient approximation. The saved float gap is the output redundancy minus the exact admissible float optimum for that instance; the quantized gap uses the exact encoded optimum.

Greater precision may distinguish more objective values while increasing signed arithmetic width and circuit work. It does not necessarily yield a monotone improvement in a stochastic, budget-limited output. Nor does a smaller optimization gap imply improved judged relevance: the objective still penalizes pair similarity subject to estimated-score constraints. This is why the primary paper reports retrieval metrics, exact objective controls, and resources together.

D. VALIDATION COVERAGE AND EXPLICIT LIMITS

Complete-iteration synthesis, including generic fixed-weight preparation and reflection, is limited to at most eight decision qubits. Native validation simulation is limited to at most five decisions. Six complete iterations passed phase-sensitive validation; 42 other simulation attempts were refused by the conservative memory guard. These refusals mark the limits of the simulations; they do not demonstrate successful execution of larger circuits. Separate small two-bit toy checks exercise nontrivial five-decision marking and reflection.

The native simulator uses a noiseless matrix-product-state method with zero configurable truncation threshold, maximum bond dimension 128, and a 10^{-8} probability tolerance. An independently constructed expected-state inverse followed by local Z expectations bounds state mismatch and workspace leakage. Every circuit worker has 120-second and 2 GiB limits. Larger native simulations, a noise sweep, hardware routing, and physical execution remain unperformed.

Construction, transpilation, native validation, compact selection, allocation profiling, and complete-pipeline timing have different boundaries. Per-method allocation profiling is a separate replay excluded from selector timing, while the fresh worker's process RSS covers its complete grid. Keeping these boundaries separate prevents an allocation replay from inflating one reported time while a costly prerequisite disappears from another.

IX. DISCUSSION, LIMITATIONS, AND REPRODUCIBILITY

A. WHAT THE COMBINED EVIDENCE ESTABLISHES

The results are consistent with an interpretation based on constrained selection. Protecting a ranking prefix produces a deterministic guarantee for the corresponding truncated ranking metric. Optimizing the tail lowers the declared pair-similarity measure for many queries, but the effect on relevance depends on errors in the neural score model and on the available candidates. The coverage and loss-distribution analyses help explain how these outcomes can occur together.

Classical-control agreement changes the appropriate attribution of the result. If constrained greedy and exact selection reach the same aggregate quality as GAS, the gain over base selection belongs to the shared candidate/objective/constraint policy. The data do not identify an extra effectiveness contribution from amplitude amplification. The small constrained domains also make exact controls practical, which is precisely why excluding them would leave the central claim weakly tested.

The cost decomposition also limits the efficiency claim: selector arithmetic accounts for a small part of the complete pipeline, while neural relevance inference accounts for a substantial share. An optimization strategy intended to improve application latency should therefore account for candidate scoring and upstream work, rather than optimize an isolated phase-query count and assume a comparable end-to-end gain. This conclusion is specific to the recorded pipeline and host; a substantially different access model would require its own measurements.

B. CONSTRUCT AND MEASUREMENT VALIDITY

Redundancy in this article is a clipped embedding-similarity average. It does not directly measure repeated facts, semantic contradiction, evidence complementarity, or answer faithfulness. Correlated but necessary evidence can receive a high penalty. Dissimilar irrelevant text can receive a low penalty. A future application-level study should test whether a chosen reduction in this proxy improves the outcomes that motivate selection in the first place.

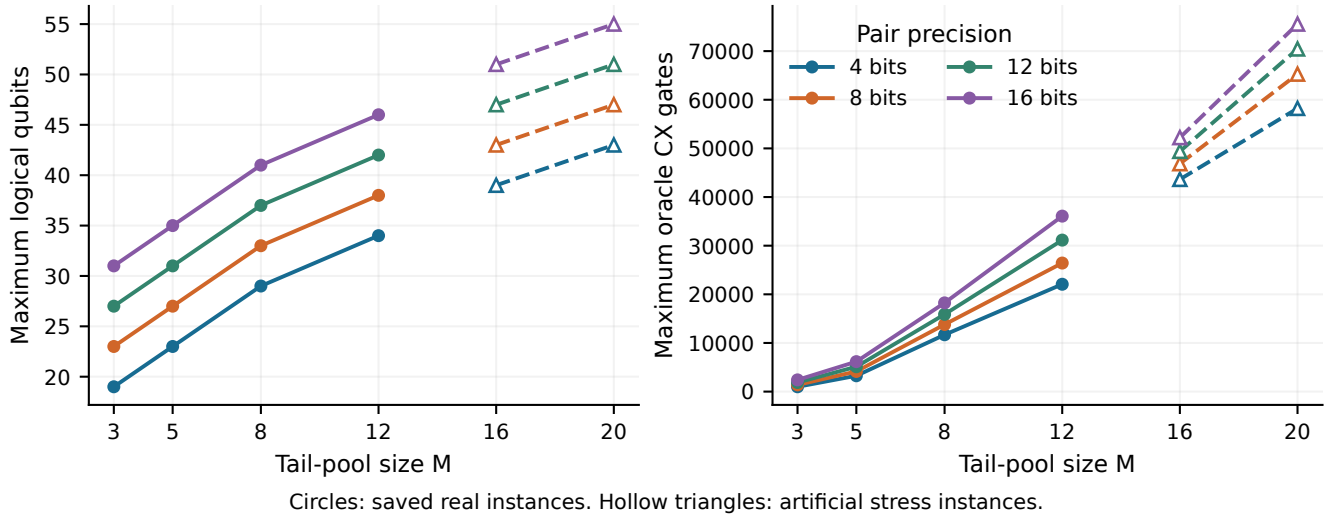


FIGURE 6. **All recorded precision levels for native logical width and oracle CX count. Real-instance points summarize the available fixed representatives; hollow triangles are artificial stress cells. Connecting lines aid inspection and are not fitted scaling models.**

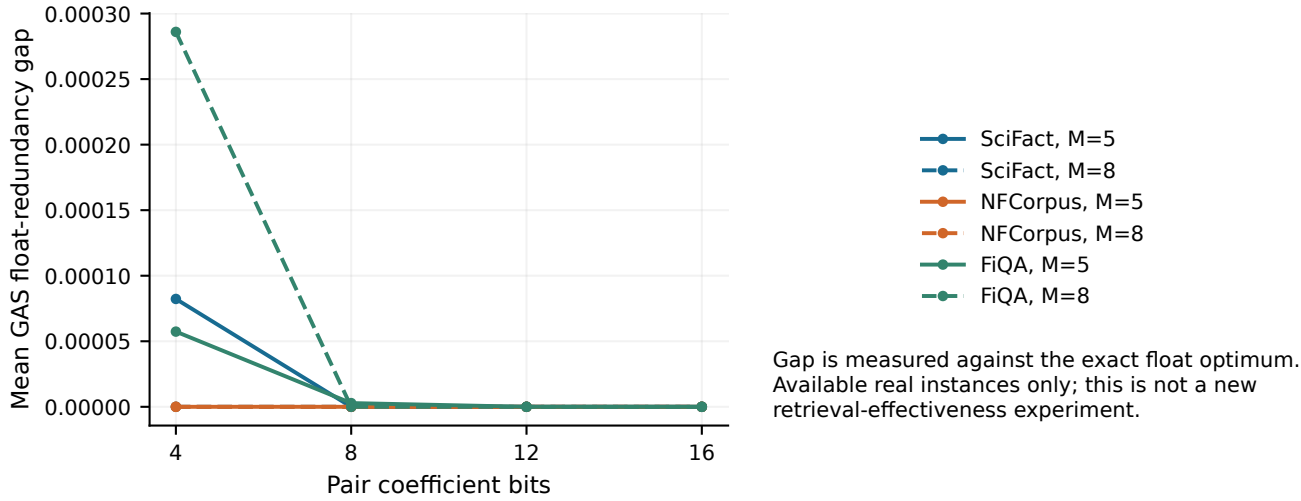


FIGURE 7. **Recorded GAS float-redundancy gaps across pair precision for available real five- and eight-candidate subpools. Values are means over the study's fixed available requests, not new retrieval runs. Search error can dominate coefficient error; the exact-optimizer regret bound does not bound these GAS curves.**

Parent-document judgments and representative passages introduce another boundary. A relevant parent may contain information outside its highest-cosine chunk, and a cross-encoder score on one representative passage may not reflect the full parent. Dense top-64 filtering, hybrid top-fifteen filtering, and score-dominance constraints can each remove useful opportunities. The journal's judgment-based ceilings reveal some of those losses, but incomplete judgments prevent treating them as ceilings on all human relevance.

Preserving nDCG@5 does not preserve every aspect of retrieval quality. Recall@8 and nDCG@8 can change, individual queries can lose substantial recall, and the tail may be used by a downstream consumer despite its lower rank. Similarly, a descriptive mean-retention gate is weaker than a prospective statistical noninferiority design. The observed failures are therefore material to the interpretation, not exceptions that can be omitted from a favorable summary.

C. INTERNAL AND EXTERNAL VALIDITY

The official tests were reused during the research trajectory. Recorded follow-up settings were not retuned on their outcomes, but prior observation limits confirmatory interpretation. NFCorpus's failed training confirmation and explicit continuation are retained. FiQA's failed refinement follow-up prevents claiming cross-domain success for the protected policy merely because its earlier base row is reported. Retrospective diagnostics and illustrative extreme cases are labeled accordingly.

The three collections cover different tasks, but they cannot

establish generalization to every domain, language, query distribution, or private collection. The measured L2/L6 controls also do not exhaust modern reranker architectures. Training-data overlap and domain mismatch can affect interpretation; exact training/test disjointness is not established by a model-family name. Prospective evaluations on new benchmarks with stronger rerankers are needed to assess these limits.

Shared-host timing and separate sessions limit latency generalization. The composition model attributes applicable upstream costs to each policy, but it is not a concurrency or production-throughput experiment. Warm-up and restoration are reported separately. No monetary, energy, quality-of-service, or physical quantum runtime measurement is available. Logical unrestricted gate counts should not be extrapolated to device performance without routing, error, and fault-tolerance assumptions.

D. ARTIFACT AND NUMERICAL REPRODUCIBILITY

The journal artifact includes the LaTeX source, tables, figures, and bibliography records. It also includes the analysis code, query diagnostics, seed metrics, and request timing components. Source mappings identify completed runs by their manifest and evaluation hashes. Export opens these runs read-only and performs no collection, selection, or neural inference. The journal scripts write only within their own artifact directory.

The verification path checks complete test-query coverage, method and seed coverage, source, model, and data identities, original-logit preservation, output cardinality, protected prefixes, and recorded feasibility. It independently recomputes 119,124 query–method–metric values from the selected saved rankings and retains the experiment-bound NIST metric checks. New coverage diagnostics additionally verify the sizes and containment of dense, hybrid, and union candidate sets. An independent request-level check verifies the additive latency identity before the stacked mean figure is generated.

The public source collections and model records are cited. Saved experimental artifacts remain local inputs to this reproducibility package and do not yet have a public archival DOI. The journal's evidence exports permit inspection of the present claims. Public deposition of the complete experiments would further support independent reproduction of the underlying runs.

E. FUTURE RESEARCH

The most direct next study would prospectively evaluate the refined policy on an untouched domain with predefined relevance and loss criteria. It should retain failed outcomes, compare against modern relevance models, and examine whether a full MMR tradeoff curve or another subset-diversity model offers a preferable operating point. New generated-answer evaluation would be needed to connect evidence-set changes to factuality, completeness, and answer usefulness.

For quantum implementation, the priorities differ: structured fixed-weight preparation, explicit reversible coefficient access, hardware routing, noise, and fault-tolerant resource estimates. These should be assessed against classical algorithms solving the same admissible objective and against the full neural retrieval pipeline. A lower logical phase count alone would not resolve the cost bottlenecks observed here.

X. CONCLUSION

We analyzed Grover-inspired evidence selection under explicit relevance, candidate, and resource constraints. The completed three-collection evidence shows that a protected five-document prefix can coexist with useful reductions in pairwise embedding similarity and with losses in judged recall. The journal extension makes those effects inspectable through candidate-pool ceilings, individual-query outcomes, seed stability, and additive request-cost analysis. It introduces no new benchmark collection or retuned policy. Matched classical optimizers attain the same reported aggregate quality on the protected test runs, while original and additional cross-encoder inference dominate their composed online costs. The separately completed reversible-oracle study exposes increasing logical resources and bounded simulation coverage, without establishing physical-device acceleration. Taken together, the findings support a carefully delimited relevance–redundancy formulation and a reproducible way to audit it. Stronger claims about generalization, generated answers, or quantum advantage require prospective effectiveness tests and independently measured implementation costs. Retaining negative outcomes and separating guaranteed prefix behavior from empirical tail quality are essential to that assessment.

APPENDIX A DETAILED QUANTIZATION AND CONSTRAINT REASONING

For the base, write every rounded coefficient as its exact scaled value plus an error in $[-1/2, 1/2]$. The total absolute error of a fixed k-element set is no greater than

$$\frac{1}{2}\left(k + \binom{k}{2}\right) = \frac{k(k+1)}{4}. \tag{24}$$

Division by $A_b k$ gives (11). This is a worst-case bound; cancellation can make the actual error much smaller. It also applies to a fixed set rather than to the outputs of two different stochastic optimization trajectories.

For refinement, let $e_A(S) = E_r(S)/(28A) - \mathcal{R}(S)$. Equation (16) states $|e_A(S)| \le 1/(2A)$. Exact optimality in the encoded objective implies

$$\mathcal{R}(S_A) + e_A(S_A) \le \mathcal{R}(S_\star) + e_A(S_\star), \tag{25}$$

$$\mathcal{R}(S_A) - \mathcal{R}(S_\star) \le e_A(S_\star) - e_A(S_A) \le 1/A. \tag{26}$$

The comparison requires the same admissible domain. A quantized optimizer operating on a different pool, a different fixed head, or different score constraints is not covered by this comparison. A GAS output that does not attain the exact encoded optimum has an additional optimization gap.

The score-count equivalence also clarifies ties. For the ascending triple $y_1 \le y_2 \le y_3$, $y_j \ge \theta_j$ holds exactly when at least $4-j$ of its values are at least θ_j. If $y_j < \theta_j$, the first j entries violate that threshold and only $3-j$ entries can meet it. The

TABLE 13. Recorded legacy development thresholds. nDCG@5 comes from saved matched-sampling outcomes. No-marked percentages count development queries, with no new ranking or test tuning.

Dataset	Threshold	Dev. nDCG@5	No-marked queries (%)
SciFact	0.3	0.6627	0.0
SciFact	0.5	0.6579	6.8
SciFact	0.7	0.6656	67.1
NFCorpus	0.3	0.3010	0.6
NFCorpus	0.5	0.3011	31.8
NFCorpus	0.7	0.3026	87.0
FiQA	0.3	0.3514	0.0
FiQA	0.5	0.3515	1.8
FiQA	0.7	0.3516	69.4

reverse implication follows by the same ordering argument. Applying this independently to each threshold proves (14) without assuming distinct logits.

APPENDIX B NATIVE VALIDATION BOUND AND REPORTING CONVENTIONS

After applying the independent expected-state inverse, the ideal joint decision/workspace state is all zero. If $p_j = (1 - \langle Z_j \rangle)/2$ denotes the probability that qubit j is one, a union bound gives

$$1 - |\langle 0 \cdots 0 | \psi \rangle|^2 \leq \sum_j p_j. \tag{27}$$

A sum restricted to workspace qubits also bounds the probability of dirty workspace. Native validation checks this bound and the norm against 10^{-8} on the small instances actually simulated. For those circuits the check is phase-sensitive. It does not validate the 42 refused simulations or other instances that were not simulated.

The saved legacy threshold analysis is retained in Table 13. The no-marked percentage is computed over queries using their recorded cosine candidates, rather than treating seeds as independent events. The development outcomes use saved matched-sampling rankings; they are not a new threshold sweep on test judgments. Lowering a threshold can reduce empty marked sets while changing the induced ranking policy, so fallback rate alone is not an effectiveness criterion.

The complete exports retain all twenty or twenty-seven measured methods, including controls omitted from compact manuscript tables. The query diagnostic CSV identifies each reference policy and candidate pool, and the seed CSV preserves request status and phase counts. A 10^{-12} tolerance is used consistently for numerical ties. The native resource CSV preserves the distinction between validated, memory-limited, unattempted, and synthetic cases. These reporting conventions prevent a missing measurement, a retained incumbent, and a successful quantum simulation from being represented as the same event.

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